

The air moved. Has exchange happened yet?

Learning intention

Explain how ventilation and alveolar structure support gas exchange, using an idealised comparison model.

Mission: separate air movement from particle movement.

Then connect both to blood transport.

What drives diffusion?

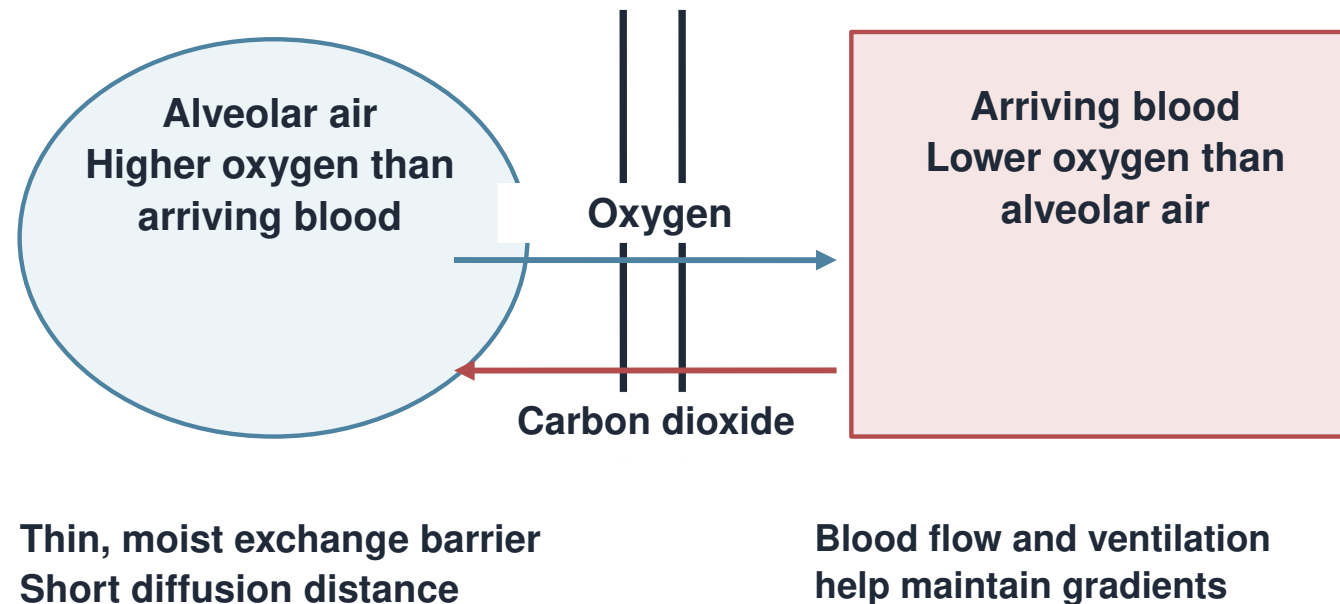
R1. In which direction is net diffusion?

R2. What does circulating blood transport from the lungs?

R3. Is respiration simply another word for breathing?

Follow one gas at a time

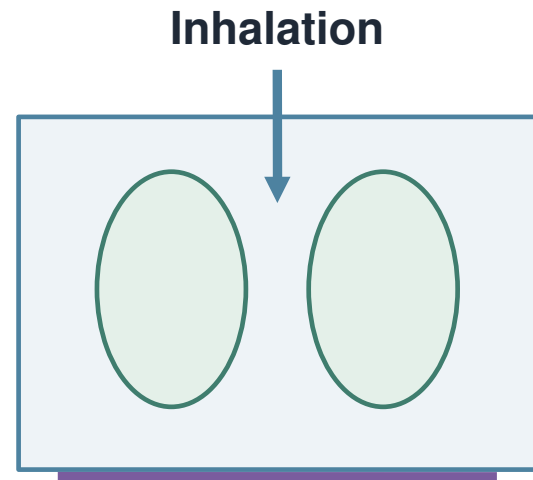
Oxygen and carbon dioxide diffuse in opposite directions



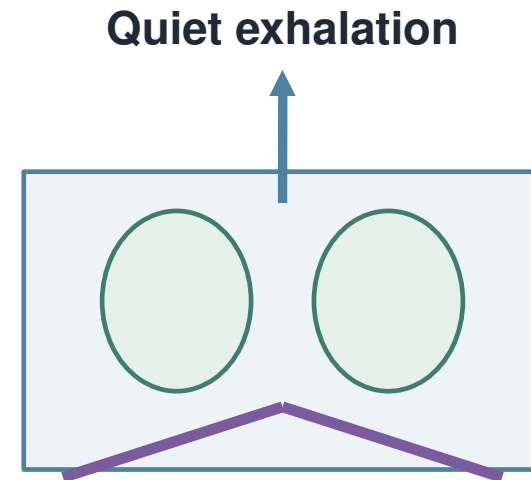
Oxygen diffuses from alveolar air into blood. Carbon dioxide diffuses from blood into alveolar air. Each gas follows its own gradient.

Air moves because pressure changes

Ventilation changes volume and pressure



**Diaphragm contracts/flattens
Volume rises; pressure falls
Air enters**



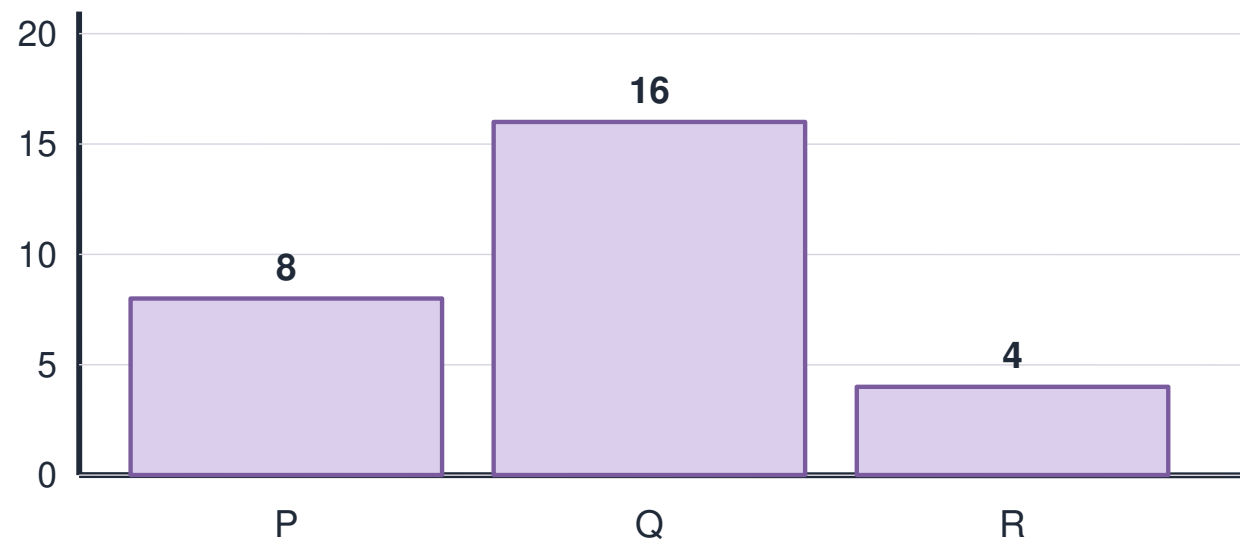
**Diaphragm relaxes; recoil
Volume falls; pressure rises
Air leaves**

Inhalation increases thoracic volume and lowers pressure. Quiet exhalation reverses the volume change.

Explain the exchange adaptations

Constructed model: compare one factor

Relative model rate index; not a clinical unit



Large area offers more exchange surface. A thin barrier gives a short diffusion distance. Ventilation and flow help sustain gradients.

Diffusion design desk

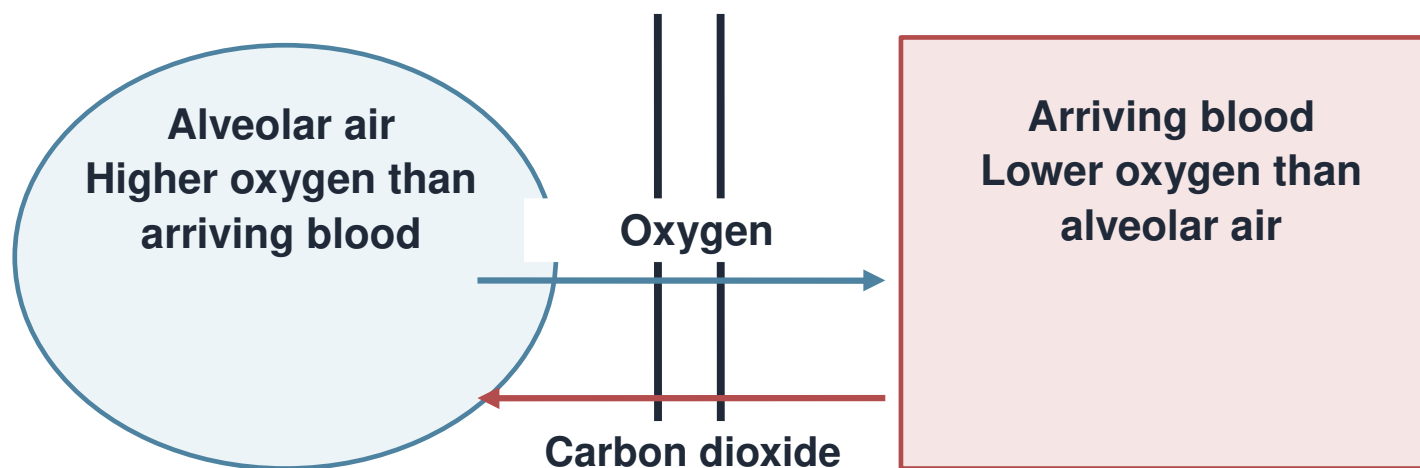
W1. Which pair changes area only? Predict the faster case. W2. Which pair changes thickness only? Predict the slower case.

Case	Relative area	Relative concentration difference	Relative thickness
P	2	4	1
Q	4	4	1
R	2	4	2

Constructed idealised comparison; rate index = area $\times$ concentration difference $\div$ thickness.

Trace a ventilation failure in the model

Oxygen and carbon dioxide diffuse in opposite directions



Thin, moist exchange barrier
Short diffusion distance

Blood flow and ventilation
help maintain gradients

W3. Fresh air stops being replaced while exchange continues. Why may the oxygen gradient decrease?

Repair a three-process caption

“The diaphragm respire oxygen into the blood.”

Replace it with ventilation, diffusion and respiration in the right places.

Which explanation names the mechanism?

Choose the supported claim.

- A** A thin exchange barrier helps because it shortens diffusion distance.
- B** The diaphragm directly pushes oxygen through blood-cell membranes.
- C** A large surface always guarantees fast exchange, regardless of every other factor.

Build an exchange explanation

Choose your route.

Use labelled arrows, the comparison table and a model limit.

Make the link from structure to function

Structure → effect on exchange → reason.

Separate model predictions from measurements.

Audience asks: what actually moves?

L1. Share one explanation.

Audience asks whether it concerns air, gas particles or blood.

Use three precise verbs

Ventilate air. Diffuse gases. Transport blood.

Then explain cellular respiration separately.

Three mechanism checks

E1. What happens to pressure during inhalation?

E2. Why does a thin barrier help diffusion?

E3. Name one process distinct from respiration.

Next useful step

Pressure • gas gradient • diffusion distance • model scope.